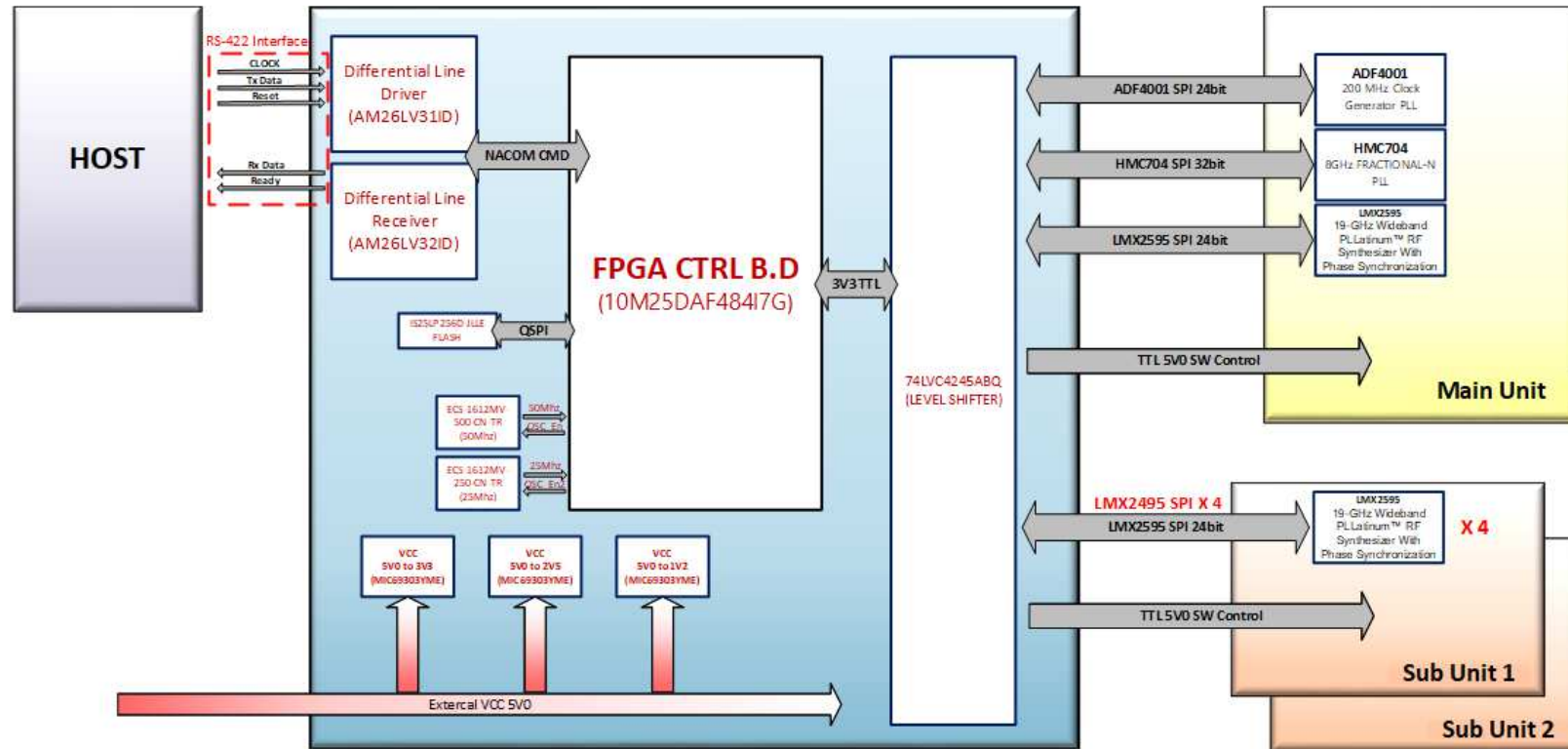
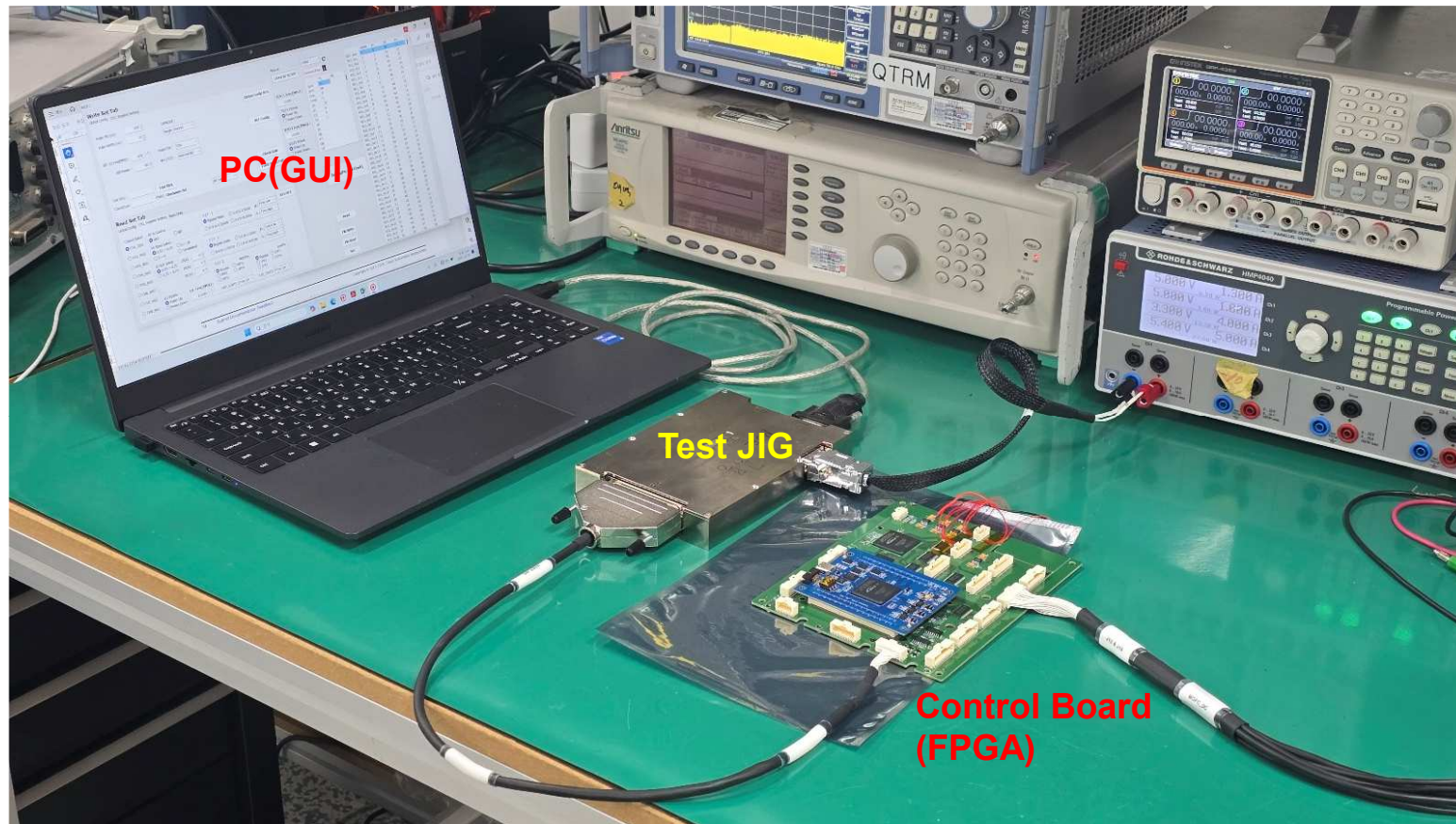


MOFC Block Diagram for Logic Circuit



MOFC Test Set up



MOFC GUI (1) – Main Path setting & Read Status

MOFC

Write Set Tab

Global Config CFG_Register Setting

Chanel Select

☒ CH1_REG

☐ CH2_REG

☐ CH3_REG

☐ CH4_REG

☐ CH5_REG

☐ CH6_REG

☐ CH7_REG

☐ CH8_REG

RF In Source

☒ ANT ☐ BIT

RF Band Select

☒ 0.35 ~ 4.75 ☐ 6 ~ 18

☐ 2 ~ 6 ☐ Terminated

IF OUT BAND

☒ 0.35 ~ 4.75 ☐ 2.75 ~ 4.75

DCA1 : 0

DCA2 : 0

DCA3 : 0

LO PDWN

☒ Power Up ☐ Power Down

LO Freq (MHz) : 10,000

T / F 1

☒ Bypass State ☐ 1.0 to 2.5GHz

☐ 0.4 to 1.25GHz ☐ 2.0 to 8.0GHz

H : Freq Low

L : Freq High

T / F 3

☒ Bypass State ☐ 1.0 to 2.5GHz

☐ 0.4 to 1.25GHz ☐ 2.0 to 8.0GHz

H : Freq Low

L : Freq High

T / F 2

HighPass

☒ Bypass ☐ HPF3

☐ HPF1 ☐ HPF4

LowPass

☒ Bypass ☐ LPF3

☐ LPF1 ☐ LPF4

HPF_STATE : Freq Low

LPF_STATE : Freq Low

REG SET

Manual

LOAD RF SETUP

COM4

Connection

TEST1 Freq (MHz) :

10,000

TEST1 PDWN

☒ Power Up ☐ Power Down

TEST LO1 Set

TEST2 Freq (MHz) :

10,000

TEST2 PDWN

☒ Power Up ☐ Power Down

TEST LO2 Set

	TX	RX
Sync		
Sync		
LenL		
LenH		
D0		
D1		
D2		
D3		
D4		
D5		
D6		
D7		
D8		
D9		

Read Set Tab

Global Config CFG_Register Setting Read Only

Temp : 0

Power Good :

Checksum STATUS : IDLE

RF RDY :

LO2 PLL LOCK

Ver-ID : 0

SERIAL : 0

[SUB2] LO8 Lock :

[SUB2] LO7 Lock :

[SUB2] LO6 Lock :

[SUB2] LO5 Lock :

ADC LOCK

[SUB1] LO4 Lock :

[SUB1] LO3 Lock :

[SUB1] LO2 Lock :

[SUB1] LO1 Lock :

BIT PLL LOCK

Status

FPGA Ver

SERIAL NUM

MOFC GUI (2) – LO Setting & Read Status

MOFC

Write Set Tab

Global Config CFG_Register Setting

Channel Select

- ☒ CH1_REG
- ☐ CH2_REG
- ☐ CH3_REG
- ☐ CH4_REG
- ☐ CH5_REG
- ☐ CH6_REG
- ☐ CH7_REG
- ☐ CH8_REG

RF In Source

- ☒ ANT
- ☐ BIT

RF Band Select

- ☒ 0.35 ~ 4.75
- ☐ 6 ~ 18
- ☐ 2 ~ 6
- ☐ Terminated

IF OUT BAND

- ☒ 0.35 ~ 4.75
- ☐ 2.75 ~ 4.75

DCA1 : 0

DCA2 : 0

DCA3 : 0

LO PDWN

- ☒ Power Up
- ☐ Power Down

LO Freq (MHz) : 10,000

T / F 1

- ☒ Bypass State
- ☐ 1.0 to 2.5GHz
- ☐ 0.4 to 1.25GHz
- ☐ 2.0 to 8.0GHz

H : Freq Low

L : Freq High

T / F 3

- ☒ Bypass State
- ☐ 1.0 to 2.5GHz
- ☐ 0.4 to 1.25GHz
- ☐ 2.0 to 8.0GHz

H : Freq Low

L : Freq High

T / F 2

HighPass

- ☒ Bypass
- ☐ HPF1
- ☐ HPF2
- ☐ HPF3
- ☐ HPF4

LowPass

- ☒ Bypass
- ☐ LPF1
- ☐ LPF2
- ☐ LPF3
- ☐ LPF4

HPF_STATE : Freq Low

LPF_STATE : Freq Low

REG SET

Read Set Tab

Global Config CFG_Register Setting Read Only

Channel Select

- ☒ CH1_REG
- ☐ CH2_REG
- ☐ CH3_REG
- ☐ CH4_REG
- ☐ CH5_REG
- ☐ CH6_REG
- ☐ CH7_REG
- ☐ CH8_REG

RF In Source

- ☒ ANT
- ☐ BIT

RF Band Select

- ☒ 0.35 ~ 4.75
- ☐ 6 ~ 18
- ☐ 2 ~ 6
- ☐ Terminated

IF OUT BAND

- ☒ 0.35 ~ 4.75
- ☐ 2.75 ~ 4.75

DCA1 : 0

DCA2 : 0

DCA3 : 0

LO PDWN

- ☒ Power Up
- ☐ Power Down

LO Freq (MHz) : 10,000

T / F 1

- ☒ Bypass State
- ☐ 1.0 to 2.5GHz
- ☐ 0.4 to 1.25GHz
- ☐ 2.0 to 8.0GHz

H : Freq Low

L : Freq High

T / F 3

- ☒ Bypass State
- ☐ 1.0 to 2.5GHz
- ☐ 0.4 to 1.25GHz
- ☐ 2.0 to 8.0GHz

H : Freq Low

L : Freq High

T / F 2

HighPass

- ☒ Bypass
- ☐ HPF1
- ☐ HPF2
- ☐ HPF3
- ☐ HPF4

LowPass

- ☒ Bypass
- ☐ LPF1
- ☐ LPF2
- ☐ LPF3
- ☐ LPF4

HPF_STATE : Freq Low

LPF_STATE : Freq Low

REG SET

Manual

LOAD RF SETUP

COM4

Connection

TEST1 Freq (MHz) : 10,000

TEST1 PDWN

- ☒ Power Up
- ☐ Power Down

TEST LO1 Set

TEST2 Freq (MHz) : 10,000

TEST2 PDWN

- ☒ Power Up
- ☐ Power Down

TEST LO2 Set

	TX	RX
Sync		
Sync		
LenL		
LenH		
D0		
D1		
D2		
D3		
D4		
D5		
D6		
D7		
D8		
D9		

MOFC GUI (2) – BIT setting & Read Status

MOFC

Write Set Tab

Global Config CFG_Register Setting

Pulse PRI (us) : 100

Pulse Width (us) : 10

BIT LO Freq(MHz) : 400

BITPower : -50

OPMODE : Single Channel

Pulse/CW : CW

INT / EXT : Internal BIT

Test REG : **Test REG**

Checksum : (Hex) **Checksum Set**

Global Config REG

BIT Config

Check SUM

LOAD RF SETUP

Read Set Tab

Global Config CFG_Register Setting Read Only

Pulse PRI (us) : 100

Pulse Width (us) : 10

BIT LO Freq(MHz) : 400

BITPower : -50

OPMODE : Single Channel

Pulse/CW : CW

INT / EXT : Internal BIT

Test REG : **Test REG**

Checksum :

Global Config REG

BIT Config

Check SUM

Manual

LOAD RF SETUP

TEST1 Freq (MHz) : 10,000

TEST1 PDWN
☒ Power Up
☐ Power Down

TEST LO1 Set

TEST2 Freq (MHz) : 10,000

TEST2 PDWN
☒ Power Up
☐ Power Down

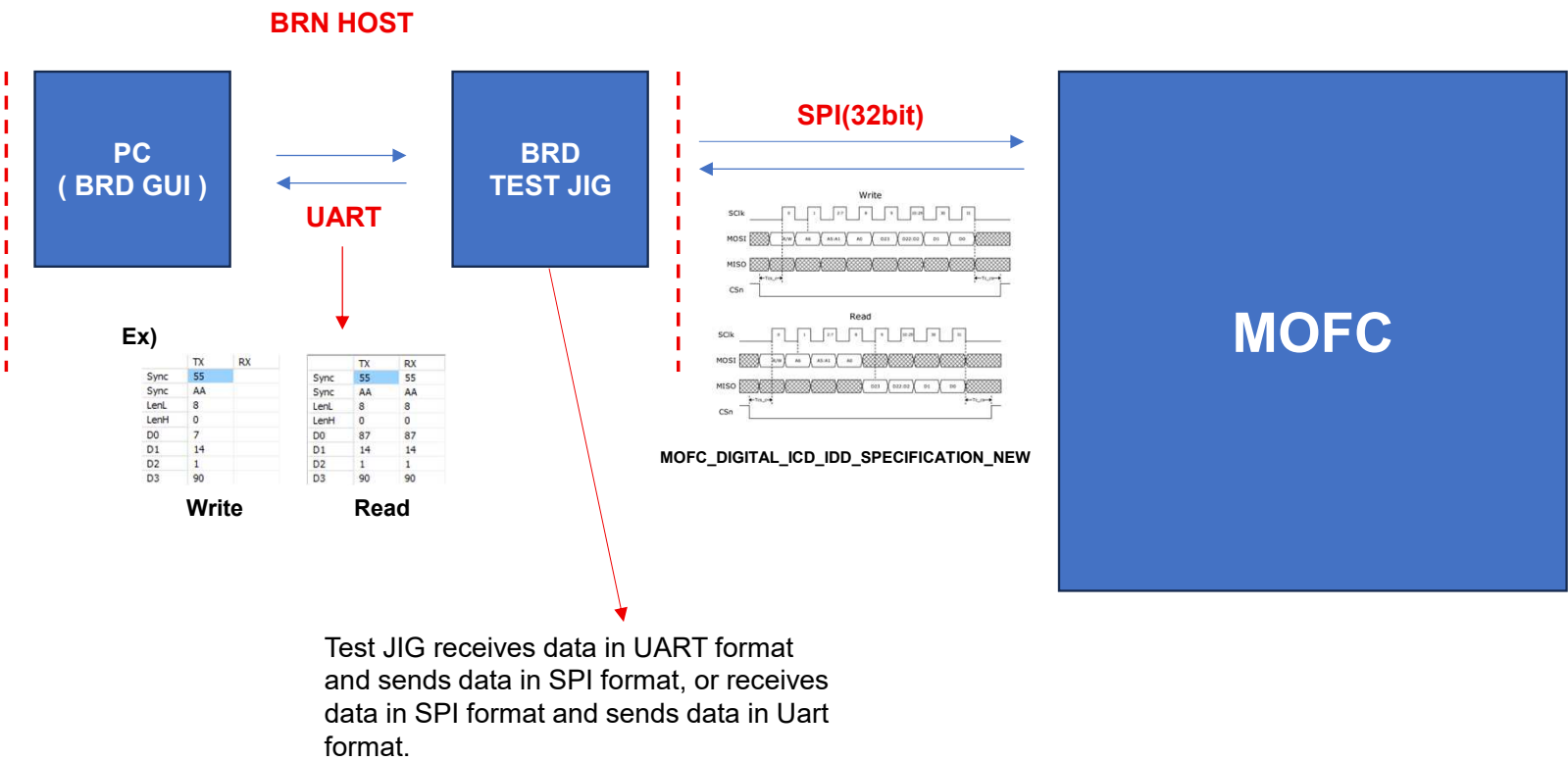
TEST LO2 Set

COM4

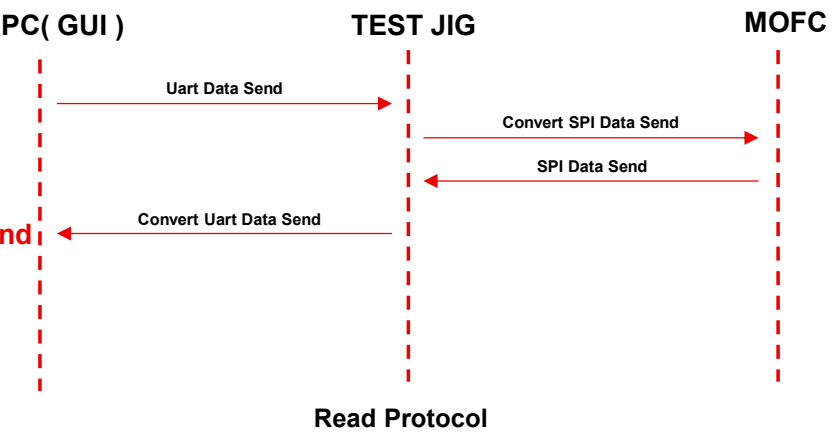
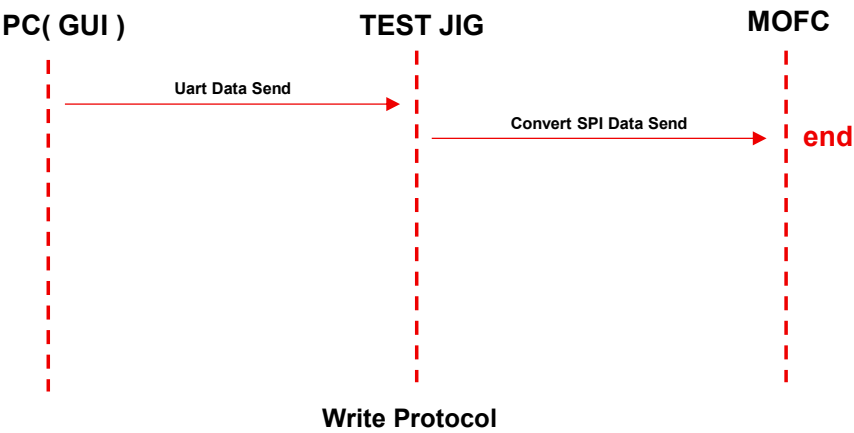
Connection

	TX	RX
Sync		
Sync		
LenL		
LenH		
D0		
D1		
D2		
D3		
D4		
D5		
D6		
D7		
D8		
D9		

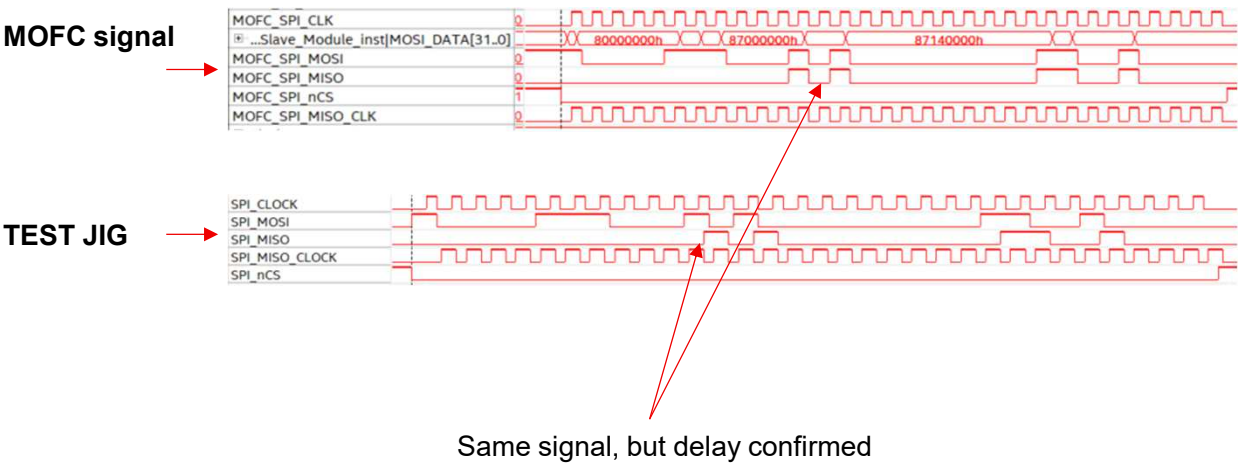
[MOFC] MOFC Communication Configuration



[MOFC] MOFC Communication Configuration



Issue : As previously noted, a hardware delay has been identified in the SPI communication. Accordingly, for read operations, adjustments are being made using the previously added SPI MISO clock while considering the nCS signal. A review is also required on the ELTA host side.



Same signal, but delay confirmed